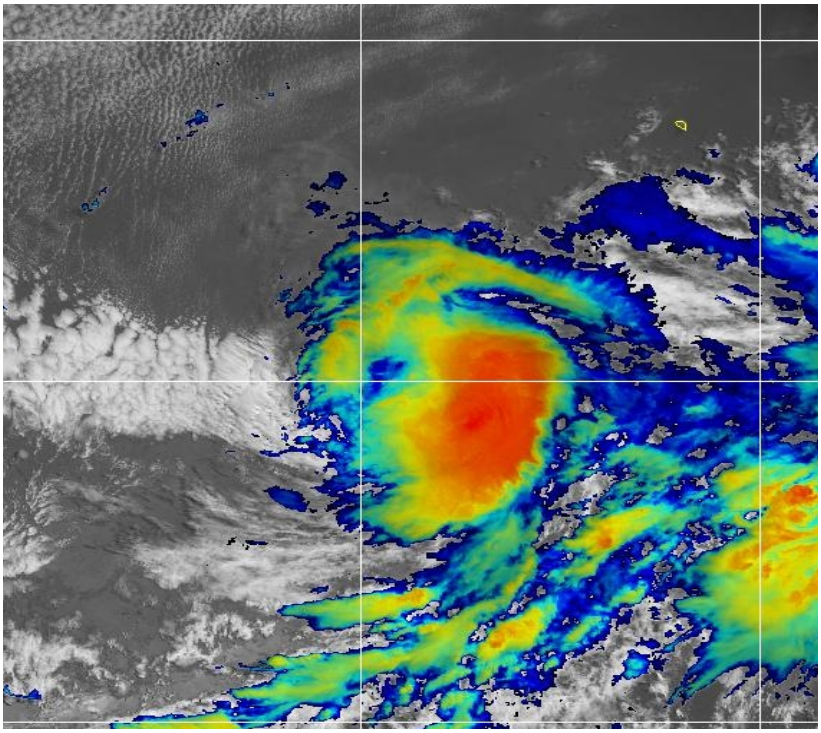
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

TROPICAL STORM COSME

(EP032025)

8–11 June 2025

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National Hurricane Center
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GOES-WEST SANDWICH RGB SATELLITE IMAGE OF TROPICAL STORM COSME AT 1800 UTC 9 JUNE 2025, NEAR THE TIME OF ITS PEAK INTENSITY. IMAGE COURTESY NOAA/NESDIS/STAR.

Cosme was a tropical storm that formed offshore of the coast of southwestern Mexico and did not bring any notable impacts to land.

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Cosme.

Tropical Storm COSME

8–11 JUNE 2025

BEST TRACK

The "best track[2]" positions and intensities for Tropical Storm Cosme are listed in Table 1. The "best track" chart of Cosme's path is given in Fig. 1, with the wind and pressure histories along with available observations[3] shown in Figs. 2 and 3, respectively. There were no credible ship observations of tropical-storm-force winds associated with Cosme.

Origin

Cosme originated from a broad area of low pressure located within the eastern North Pacific monsoon trough about 500 n mi offshore of southwestern Mexico on 5 June. A tropical wave that emerged from the west coast of Africa on 24 May and crossed Central America on 2–3 June may have also contributed to its genesis. Showers and thunderstorms associated with this system gradually became better organized as it began to lift north-northwestward on 6–7 June. The system is estimated to have become a tropical depression at 0600 UTC 8 June, when deep convection consolidated over the low-level circulation center, with curved bands wrapping into the center from the west and southern semicircles. The cyclone strengthened into a 35-kt tropical storm about 6 h later.

Peak Intensity and Minimum Pressure

Cosme's estimated peak intensity of 60 kt from 1800 UTC 9 June to 1200 UTC 10 June is supported by subjective Dvorak estimates from TAFB and SAB, along with objective estimates from UW-CIMSS. Ultra-high-resolution ASCAT data during this period (Fig. 4) showed peak winds of 45–50 kt, while an RCM-1 synthetic aperture radar (SAR) pass from a few hours earlier indicated maximum winds of 63 kt in the southeastern quadrant. Although offset by a few hours, the two passes revealed a similar wind structure. However, it is worth noting that SAR wind data

[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Cosme.

are still being evaluated for their application in estimating peak winds and structure in tropical cyclones.

The estimated minimum central pressure of 991 mb is a blend of the Knaff-Zehr-Courtney (KZC) pressure-wind relationship and subjective and objective Dvorak estimates.

CASUALTY AND DAMAGE STATISTICS

There were no reports of damage or casualties associated with Cosme.

FORECAST AND WARNING VERIFICATION

Table 2 provides the number of hours in advance of formation with the first NHC Tropical Weather Outlook (TWO) forecast in each likelihood category. NHC introduced the feature in the TWO in the low category (<30%) in both the 2- and 7-day periods, 54 and 60 h before genesis, respectively, although the 7-day outlook provided much shorter-than-average lead time. This reduced lead time was partly due to the proximity of Tropical Storm Barbara, which also formed several hundred miles to the northeast of Cosme at 0600 UTC 8 June. Its presence added uncertainty in the numerical guidance, as the models struggled to depict both systems until shortly before formation. Probabilities reached the high category (>60%) within 24 hours of the system's initial inclusion in the TWO. Fig. 5 shows composites of 7-day TWO genesis areas for each category prior to Cosme's formation. While the genesis location was captured within all of the outlook areas, it ultimately occurred near the southern edge of most of the regions depicted.

A verification of NHC official track forecasts for Cosme is given in Table 3a. Official track forecast errors were lower than the mean official errors for the previous 5-year period at 12, 24, and 60 hours. At 36 and 48 hours, the forecast errors were higher than the 5-year means, although the small sample size limits meaningful interpretation. A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b.

A verification of NHC official intensity forecasts for Cosme is given in Table 4a. Official intensity forecast errors were lower than the mean official errors for the previous 5-year period at all forecast times through 60 hours. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b.

There were no coastal watches or warnings issued for Cosme.

ACKNOWLEDGEMENTS

Dr. Philippe Papin (NHC) created the Graphical TWO verification figure.

Table 1. Best track for Tropical Storm Cosme, 8–11 June 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
08 / 0600	11.7	110.0	1005	30	tropical depression
08 / 1200	12.3	110.4	1002	35	tropical storm
08 / 1800	13.0	110.8	1000	40	“
09 / 0000	13.5	111.6	1000	40	“
09 / 0600	14.0	112.4	999	45	“
09 / 1200	14.4	113.0	994	55	“
09 / 1800	14.7	113.6	991	60	“
10 / 0000	15.0	114.1	991	60	“
10 / 0600	15.1	114.5	991	60	“
10 / 1200	15.3	114.9	993	60	“
10 / 1800	15.6	115.2	999	50	“
11 / 0000	15.9	115.0	1001	45	“
11 / 0600	16.0	114.5	1004	40	low
11 / 1200	16.2	113.8	1006	30	“
11 / 1800	16.5	112.9	1008	25	“
12 / 0000					dissipated
09 / 1800	14.7	113.6	991	60	minimum pressure & maximum winds

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	54	60
Medium (40%-60%)	42	54
High (>60%)	30	42

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Tropical Storm Cosme, 8–11 June 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**19.0**	**26.7**	46.2	69.6	**64.6**			
OCD5	27.8	65.2	84.4	77.7	145.5			
Forecasts	9	7	5	3	1			
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4	108.6	135.2
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0	340.7	422.7

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Tropical Storm Cosme, 8–11 June 2025. Errors smaller than the NHC official forecast are shown in boldface type.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	19.0	26.7	46.2	69.6	64.6			
OCD5	27.8	65.2	84.4	77.7	145.5			
GFSI	32.9	60.6	93.5	140.6	232.3			
HMNI	32.8	59.1	89.8	128.0	235.8			
HWFI	31.8	50.4	76.0	107.0	**29.6**			
EMXI	**13.9**	**25.9**	49.7	**48.8**	**21.4**			
CMCI	52.5	103.3	137.6	119.2	105.1			
AEMI	34.7	76.6	128.3	175.2	188.3			
HCCA	19.5	28.3	**39.8**	**48.4**	**47.0**			
FSSE	**17.8**	28.7	47.4	**58.3**	98.9			
TVCX	21.4	35.9	47.0	69.9	111.0			
GFEX	**18.8**	29.4	**45.5**	**54.9**	122.6			
TVCE	22.0	35.2	50.8	78.5	112.1			
TVDG	20.2	34.2	52.3	79.2	128.1			
TABD	30.9	70.0	113.7	156.1	212.0			
TABM	25.2	60.9	103.4	156.2	222.1			
TABS	30.2	85.6	157.1	222.2	281.6			
Forecasts	9	7	5	3	1			

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Tropical Storm Cosme, 8–11 June 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**3.3**	**3.6**	**4.0**	**5.0**	**5.0**			
OCD5	5.4	11.3	9.8	8.0	5.0			
Forecasts	9	7	5	3	1			
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5	17.0	18.6
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3	21.5	21.6

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Tropical Storm Cosme, 8–11 June 2025. Errors smaller than the NHC official forecast are shown in boldface type.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	3.3	3.6	4.0	5.0	5.0			
OCD5	5.4	11.3	9.8	8.0	5.0			
HWFI	5.0	7.9	8.4	5.0	8.0			
HMNI	4.2	5.3	5.6	**3.3**	**3.0**			
HFAI	4.3	6.1	7.2	7.0	11.0			
HFBI	5.4	11.0	16.2	13.7	9.0			
DSHP	5.4	10.4	13.6	11.0	**1.0**			
LGEM	5.3	11.9	15.4	12.7	13.0			
ICON	4.2	8.0	10.0	6.3	**2.0**			
IVCN	3.3	5.0	5.4	**3.3**	**2.0**			
IVDR	**3.2**	4.3	4.4	**3.7**	**4.0**			
GFSI	5.6	10.0	9.2	7.0	6.0			
EMXI	4.9	8.9	13.2	18.3	18.0			
HCCA	3.9	6.3	9.0	7.3	5.0			
FSSE	**3.2**	3.9	4.6	**4.0**	6.0			
Forecasts	9	7	5	3	1			

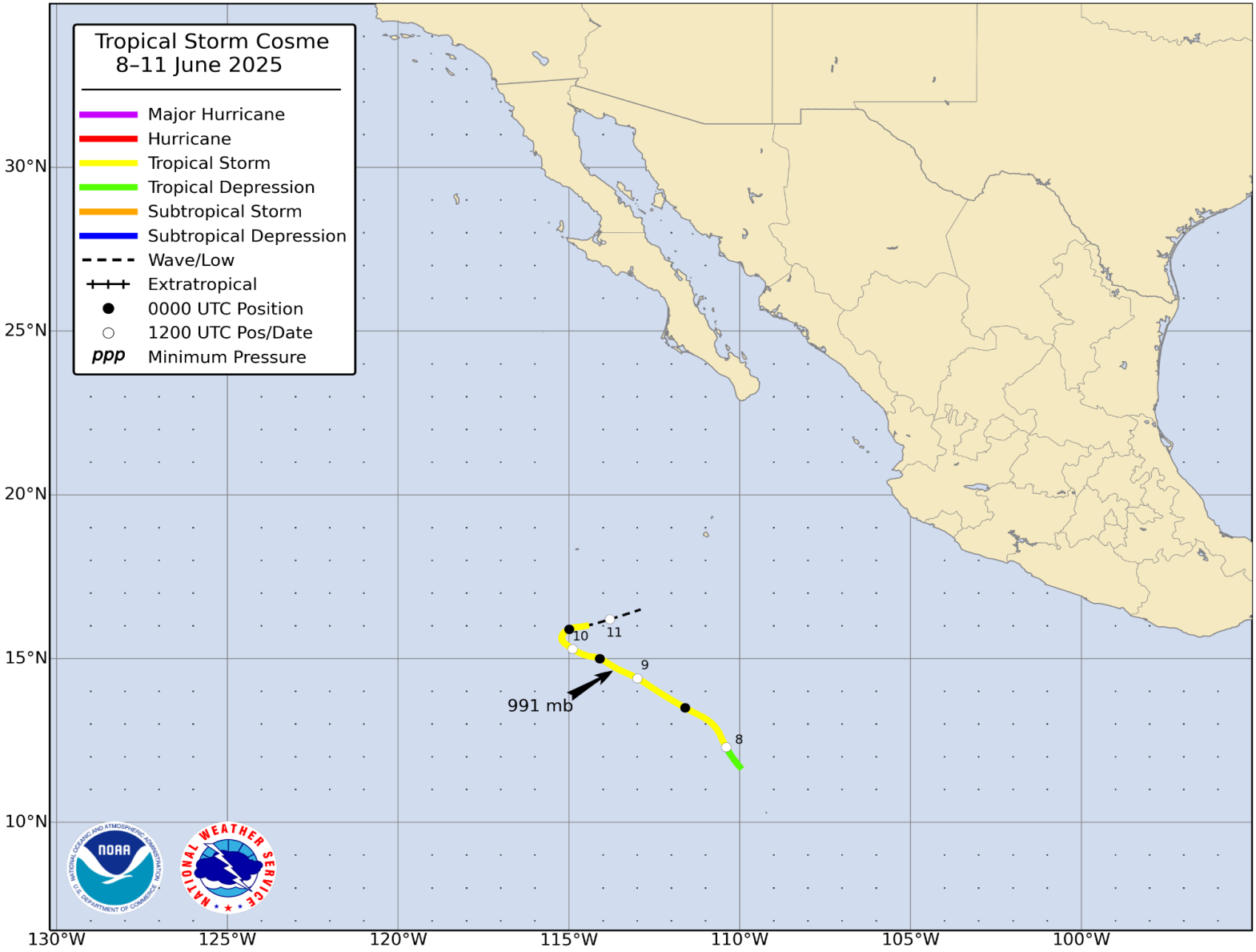


Figure 1. Best track positions for Tropical Storm Cosme, 8–11 June 2025.

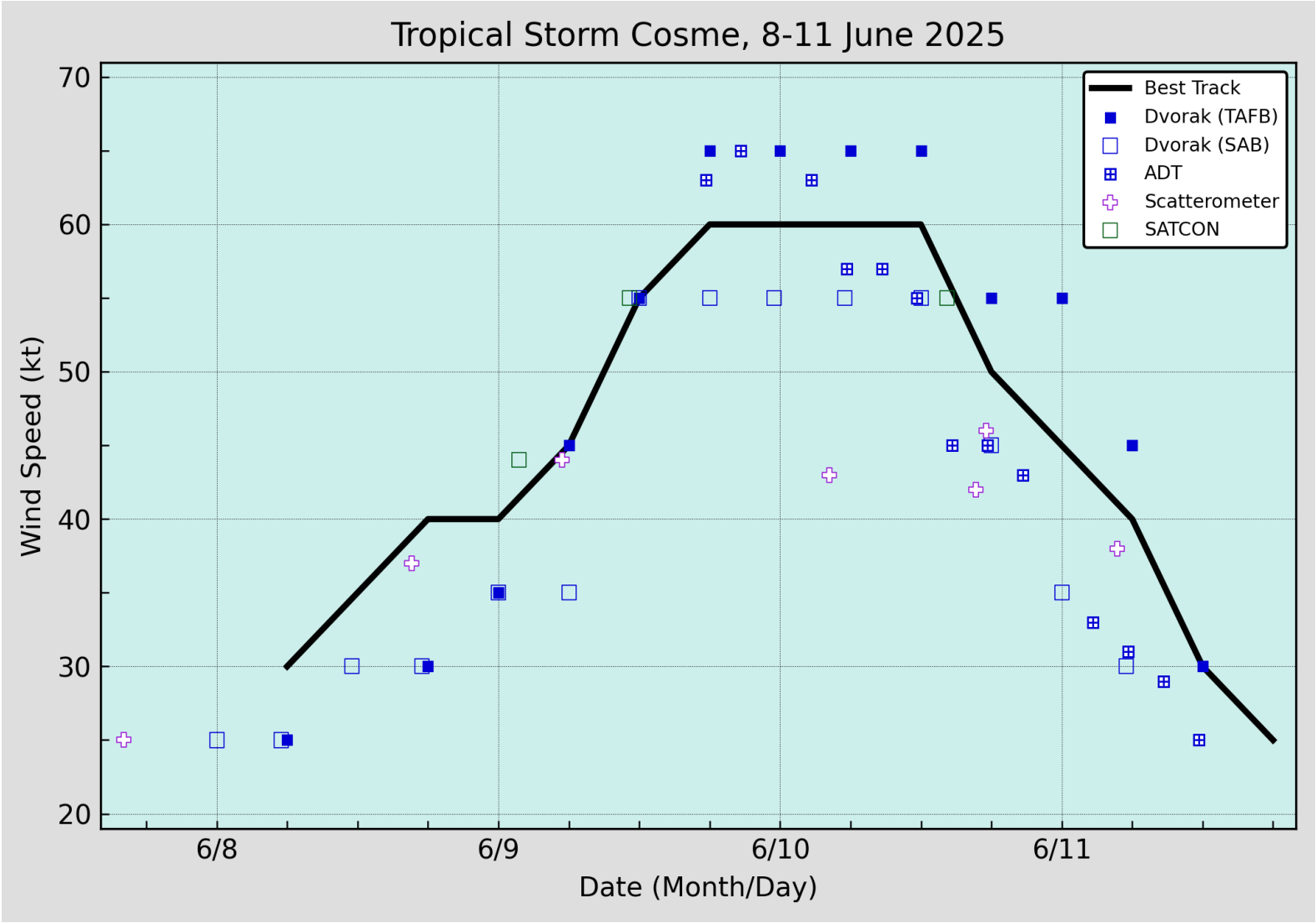


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Tropical Storm Cosme, 8–11 June 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

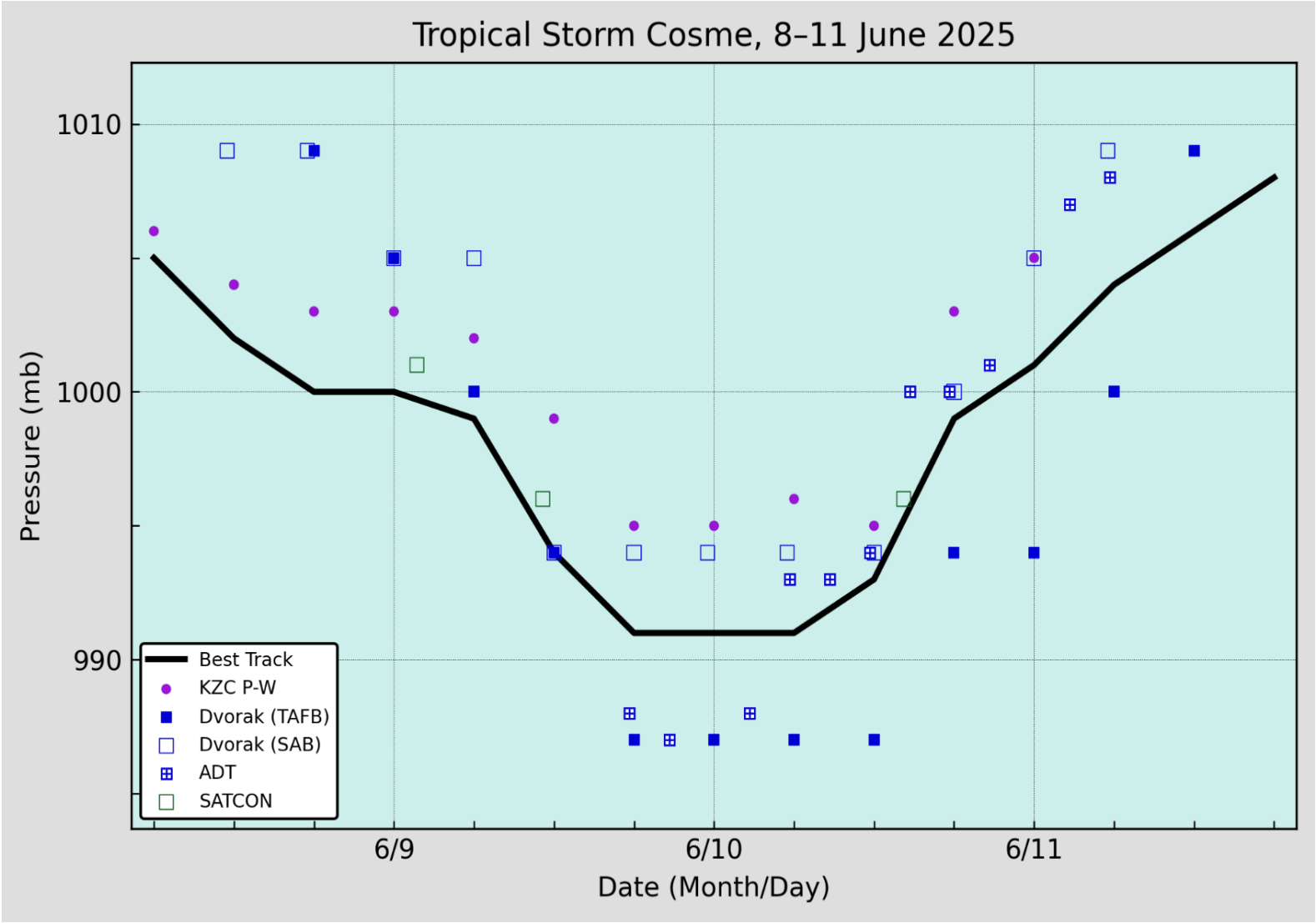


Figure 3. Selected pressure observations and best track minimum central pressure curve for Tropical Storm Cosme, 8–11 June 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

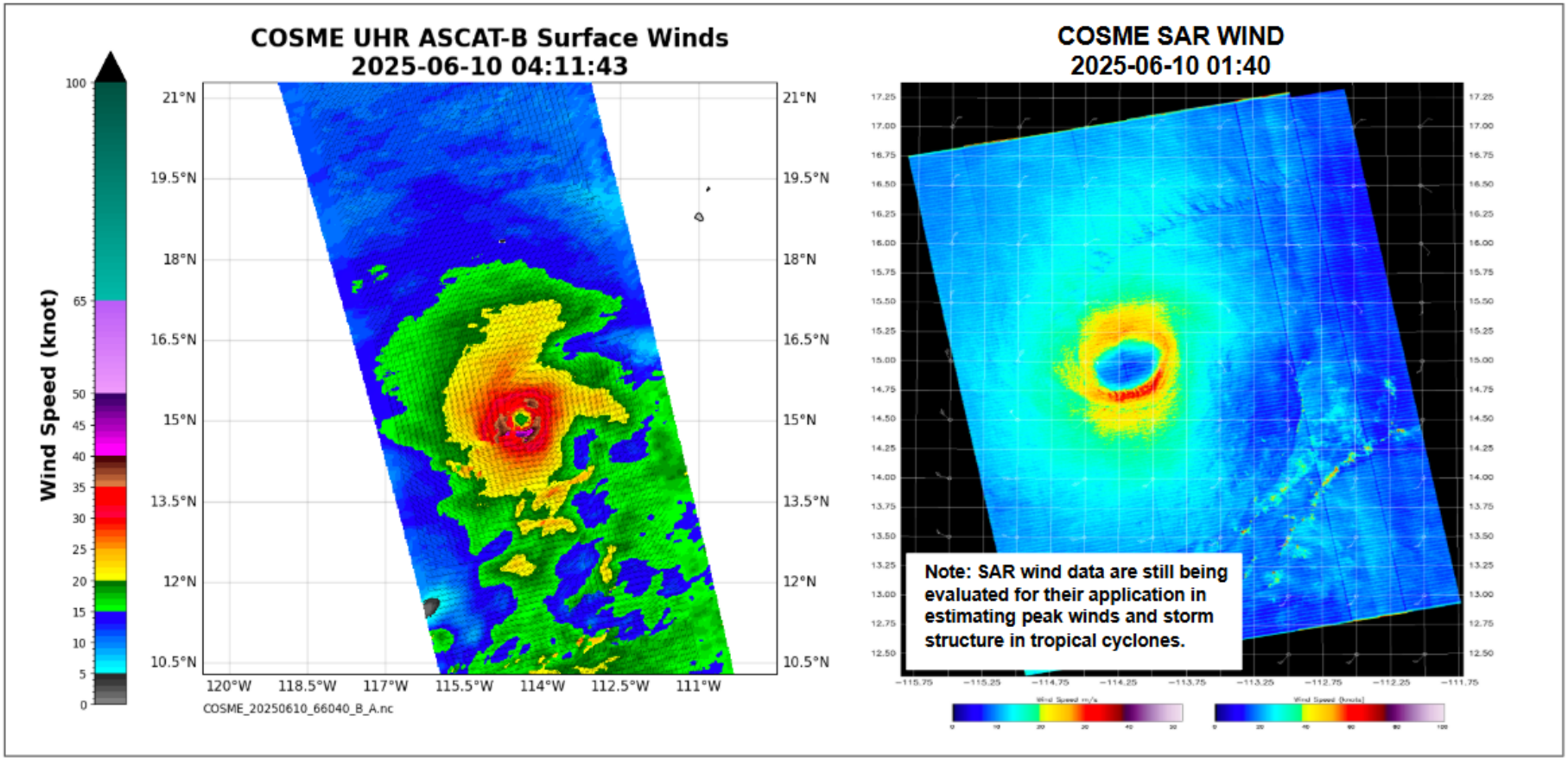


Figure 4. Surface winds from an Ultra-High Resolution (UHR) ASCAT-B pass at 0411 UTC 10 June (left) and from an RCM-1 synthetic aperture radar (SAR) pass at 0140 UTC 10 June (right). ASCAT indicated peak winds of 45–50 kt (23–26 ms^{-1}), while SAR showed maximum winds of about 63 kt (32 ms^{-1}) in the southeastern quadrant. Although offset in time by a few hours, the two passes reveal a similar wind structure; however, as noted in the figure, SAR wind data are still being evaluated for their application in estimating peak winds and structure in tropical cyclones. Data courtesy of NOAA/NESDIS/STAR Ocean Surface Winds Team.

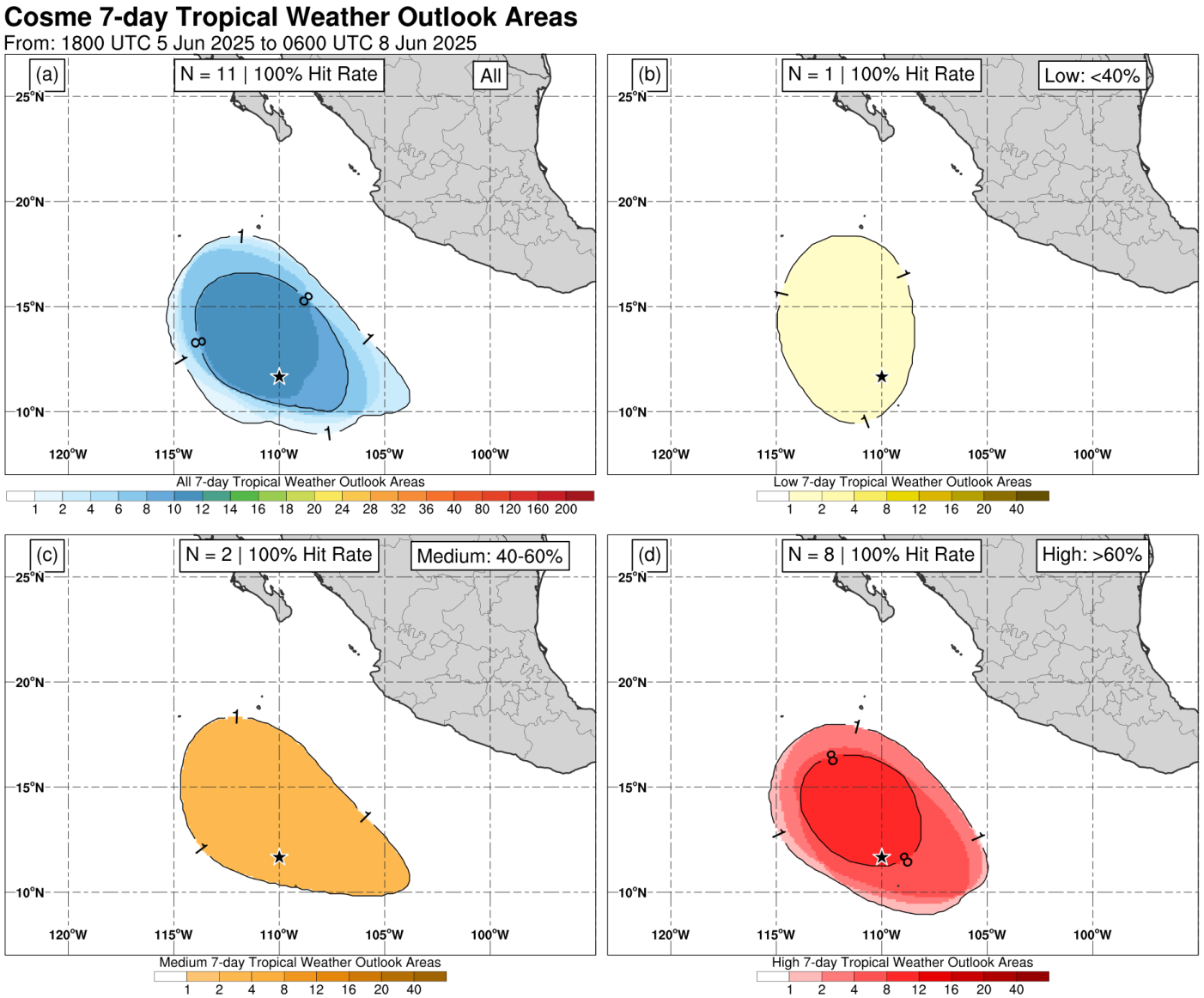


Figure 4. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Cosme for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.